Rising Waters

Flood Management System — AI/ML Project

Coding & Solution

Phase 5: Project Development Phase

IBM Cloud | XGBoost 96.55% | Skill Wallet | AI/ML Track

ML Pipeline Implementation

The model training pipeline follows a 7-step process as specified in the project scope:

Step 1	Environment Setup — Python, NumPy, Pandas, Scikit-learn, XGBoost, Flask, Joblib
Step 2	Dataset Collection — 5,000 rows, 6 meteorological features, binary target
Step 3	Data Visualisation — 4 EDA charts (distributions, box plots, pairplot, heatmap)
Step 4	Preprocessing — Mean imputation, IQR outlier capping, StandardScaler
Step 5	Model Building — Decision Tree, Random Forest, KNN, XGBoost
Step 6	Best Model Selection — XGBoost (96.55% accuracy), saved as floods.save
Step 7	Flask App — 6 HTML pages, 2 dedicated result pages, prediction history

Flask Application Routes

GET /	Home Dashboard — model stats, prediction counts
GET /predict	Input Form — 6 meteorological parameter fields
POST /predict	Process form → validate → scale → predict → redirect
GET /result/flood	Flood Chance Result — red banner, emergency protocol
GET /result/noflood	No Flood Result — green banner, monitoring actions
GET /history	Prediction audit log from predictions_log.csv
GET /about	Project architecture, model info, use case scenarios